

Image Classification of Corn Seeds using Ensemble-based CNN

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Abstract— As corn has become staple food for people in many countries around the world, that directly increases its production. The versatile nature of maize plant is the reason for its highest cultivation among all other staple foods. As many food industries demand for good quality of corn seeds for usage in their own products. The quality of corn seeds depends on the various factors like type, color, condition, texture, size, etc. The image classification technique using Convolutional Neural Networks shown high accuracy in the selection of good quality of corn seeds. But on the other hand, for image classification of tons of corn seeds, CNN has to extract large number of features. Therefore, this paper proposes the image classification of corn seed using the CNN and ensemble learning technique. The proposed work shown good performance compared to the conventional CNN methods.

Keywords—CNN, Image Classification, ensemble learning, corn seed, corn kernel

I. INTRODUCTION

Maize is the largest grown feed crop worldwide, which has become the staple food at many places, increasing its production range. The United States contributes the maximum production [1] of Maize in the world where it producing 387 billion metric tons of corn in total of 1147 million tons in the year 2023/2024. It is not only used as a food for humans and livestock but also as a crude material and fuel in industries. The versatile characteristics of maize and its adaptability to weather conditions help the farmers to cultivate it throughout the year at ease. Corn is the food harvested product of the maize.

Demand for corn increases mainly because of its high consumption. It is consumed as raw and many food industries are utilizing the benefits of corn for producing more food products based on it. Henceforth the importance of high-quality selection of corn becomes the main motivation in the current research. The human visualization of corn seeds is the best way to select high-quality seeds. But in practice, for billion tons of corn seed selection, it reduces the possibility and consumes more time. Therefore computer-aided visualization [2] of corn seeds is used to obtain seeds with good characteristics. Image processing plays vital role in the prediction of high-quality corn seed based on its texture, condition, type, colour, shape, etc. Image processing technology depends on Artificial Intelligence (AI) and Machine Learning (ML) algorithms in various steps of it like, image acquisition, enhancement, restoration, feature extraction, segmentation, classification, prediction, etc. Of many techniques employed, the Convolution Neural Network (CNN) shows outstanding performance in producing top-grade outputs. Therefore, many researches carried out in the selection of corn seeds using image processing employing the CNN. The CNN has many layers embedded which processes the image considering its properties at different stages and

produces the accurate outputs. The CNN accuracy increases by introducing many layers for convolution processes. In the paper [3], the maize seed classification is done by deep feature selection and fusion using CNN models considering the MRMR (Max Relevance and Min- Redundancy) method for selecting the features of maize seed that is trained and tested, then it undergoes the fusion processes utilizing the CNN algorithm for haploid-diploid classification of it. Hyperspectral imaging is one of the non-invasive and non-destructive method for collecting and processing the images in the electromagnetic spectrum field. It produces more accurate output at rapid rate. The authors in the paper [4] uses hyperspectral imaging of corn seed and used DCNN, SVM (Support Vector Machine), and KNN (K Nearest neighbour) algorithms, where they concluded DCNN is most prominent one. In the objective of framing lost model, CNN based on transfer learning is proposed in the paper [6] where the results are simulated for AlexNET, GoogleNET, VGGNet, and ResNet for their own datasets and proved VGG-19 shown most accurate results. The edges of the corn kernel are given more importance using Convex Hull in the CNN that reduces the effects of background and data augmentation is carried out using six classes that overcomes the issue of overfitting id the technique used in the paper [7]. The defect and defect free corn seeds are detected from the RGB and near infrared images capture using the multispectral camera is explained in the paper [10] and applied two pathway CNN with the benefits of the Watershed algorithm is proposed that results in high quality throughput control. In the paper [11], the number of corn kernels present in its ear are counted using the CNN learning systems using the sliding window algorithm where the overlapping outputs were reduced using the non-maximum suppression method. At last stage, the centre of corn kernel patches are detected using the CNN regression method. The paper [12] provided a vast survey on food grain classification systems. Corn seed vitality and Vigor is explained in the paper [13] the hyperspectral images and one-dimensional CNN is used to choose the best one from 144 samples is carried out and the same with two dimensional CNN shown 99.96% accuracy. The requirement of seed qualities like viability, Vigor, fungus formation detection, damage detection, grading, cleanness detection, determination of seed composition, etc can be examined with the help of hyperspectral images that help in the analysis of the number of samples, spectral range, features of an object, condition of a seed, status, orientation of the sees, composition etc. [14]. These images with the help of machine learning techniques make greater impact in the study and detection of the seed quality. The hyperspectral images of maize seed are acquainted and classified using successive projections algorithm where the spectral imaging features extracted and combined with feature transformation methods using Least Square Support Vector machine (LS-SVM) is proposed and

proved with good accuracy in the paper [15]. To overcome the issue of limited training samples present, transfer learning algorithm is used with the CNN for classification purposes where the ensemble data are used for learning and applied to CNN layers for image processing purposes is proposed in the paper [19]. The authors in the paper [21] used Extreme learning Machine (ELM) and CNN together in the image classification process cause CNN outperform in feature extraction and ELM classifies the images at faster rate, therefore after extracting features of images with CNN it is applied to ELM where the classification results in good accuracy.

According to the survey, corn seeds require a high level of classification in terms of colour, shape, quality, condition, etc. for large quantities of production. This paper proposes the novel technique of using an ensemble learning algorithm with CNN in the image classification of corn seeds. The paper is organized as follows, i) Introduction, ii) Deep Models in Image Processing, iii) Image processing of corn seeds, iv) Proposed work, v) Results and Discussion, and vi) Conclusion.

II. DEEP LEARNING MODELS IN IMAGE PROCESSING

A. Convolutional Neural Network

CNN [18] plays a vital role in image processing applications that help the system to study the visual data in depth. Convolution is the mathematical operation involved in the layers of a CNN network. Each layer depends on the output of its predecessor layers. Every layer with the help of extracted data from the previous layers, extracts the features of the layer in depth which makes the CNN [16] deep learning model a successful one in the image processing field. There are three layers associated with it,

- The Convolutional Layer, it is the main layer of CNN where the pixels in the images are filtered by sliding concept using the width and height of the pixel and perform the dot product using convolution transform, which results in the feature map. Thus, it contains many stacked layers where the raw image data is converted into varieties of feature maps where each map gives detailed information about a information like edge, textures, colors, scenes, objects, etc.
- Pooling Layers, it is layer for down sampling the information obtained from the convolutional layer. Its main objective is to reduce the data for processing. It reduces the data size either by averaging or maximum pooling method. But in another way it produces lossy information content.
- Fully connected layers, as the name indicates it summarizes all the features extracted and classify the images accordingly. The continuity is formed by this layer by connecting the previous neuron to the present neurons.

In image processing, CNN finds it application in the areas of image classification, feature extraction, segmentation and image classifications.

B. Ensemble Learning Algorithm

Ensemble learning-based deep learning [17] predicts accurate results by dividing an image into different features

based on the feature selected, variance and bias are reduced. It is of three types, bagging, boosting, and stacking.

- Bagging, also called bootstrap-aggregation or parallel ensemble modelling, where the process of prediction takes place in a parallel manner considering various features of image for fitting the input data.
- Boosting method is called a sequential ensemble learning here the additive process will take place in each steps, where the previous step considers a feature following the next levels.
- Stacking is called heterogeneous ensemble learning where the heterogeneous data are taken for predicting a feature in an image.

Ensemble learning [20] in the image processing application helps to obtain accurate results with the help of training models and also increasing the levels increase the percentage of accuracy.

III. IMAGE PROCESSING OF CORN SEEDS USING CNN

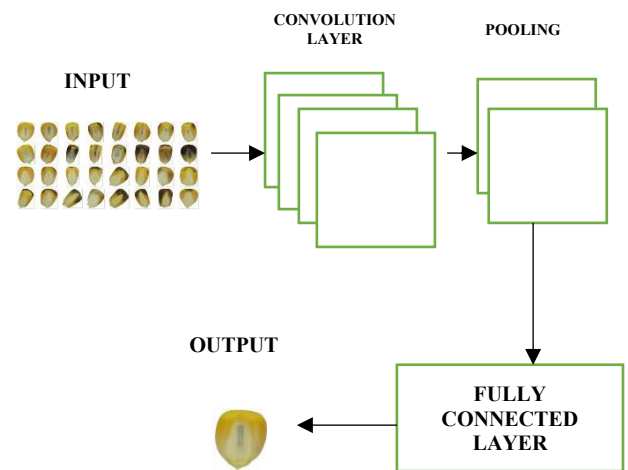


Fig. 1. CNN processes for Corn seed processing

As many of the food industry depends on the corn seed production, it should determine the corn properties [24] for handling, separating, storing and processing it. The CNN technique [5] can be used in the classification process of corn seed in terms of color, shape, texture and varieties. CNN shown good result in the corn seed classification and feature extraction when applied with different datasets of seeds and using different capturing technologies [8]. The Fig.1. shows the CNN processing of corn seed to obtain the accurate output. After preprocessing, the images of corn seed are sent to CNN model, where it undergoes convolution layers. The convolutional layers obtain the features and reduce the dimension in the pooling layer. Using Fully connected layers, the features are tested and output is predicted. Based on breeding duration, the corn seeds are classified as haploid and diploid. Haploid contains a single set of chromosomes that fastens the breeding generation in two to three years. In diploid corn seeds, it takes eight years for a generation with 20 chromosomes that avail a set of chromosomes form each parent. This classification can also be carried out using the CNN model by training the convolution layers to extract the

features related to chromosome properties. Other parameters like color [22], texture, shape, etc. can be predicted using available datasets. CNN touched the greater accuracy of the prediction of the corn seeds [9]. It is most important to maintain the quality and characteristics of corn seed [23] when it is stored for a long duration by food industries.

IV. PROPOSED MODEL

From the survey, it is clear that the CNN model classifies the corn seed based on various types and properties. But when it comes to corn seeds images containing more amount, the CNN requires more convolutional layers for extracting the features that increases the complexity of the system. Therefore, this paper proposes the incorporation of ensemble learning deep learning technique with CNN for image classification of corn seeds. The proposed model is presented in the Fig. 2. The corn kernel images are pre-processed and classified using CNN model. After the classification using CNN, the ensemble learning model is used to obtain the accurate result based on the requirements.

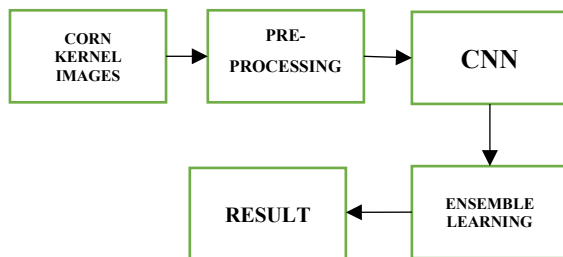


Fig.2. Proposed Model

The quality of corn seed depends on properties like color, texture, shape, size, and type of it. The CNN efficiently classifies these properties when the number of layers is increased. In order to maintain the nominal computation level, the ensemble learning model is added which takes the features obtained from the CNN and ensemble it into different groups. Then the groups separately perform training and testing based on the feature named as bagging ensemble model. This paper uses bagging ensemble learning because it is faster as the system parallelly predicts the different features of the corn kernel seed. It is a more efficient method where the results are aggregated in the last and predicts the output. In this paper, the ensemble group are framed for color, shape, size, texture and type. Based on the tested data obtained from the ensemble groups, the results will be aggregated. The ensemble learning mainly depends on the variance parameter. If the result aggregated is below the minimum variance value the corn kernel seed will be selected otherwise it will be rejected. As the training data set contains perfect feature value, therefore the minimum variance indicates that the particular corn kernel seed maintains the maximum properties of the seed that can be selected for different usage. Fig. 3. depicts the flow diagram of the proposed model using CNN and ensemble learning for corn kernel seeds. Here the output from the CNN model is considered to the feature

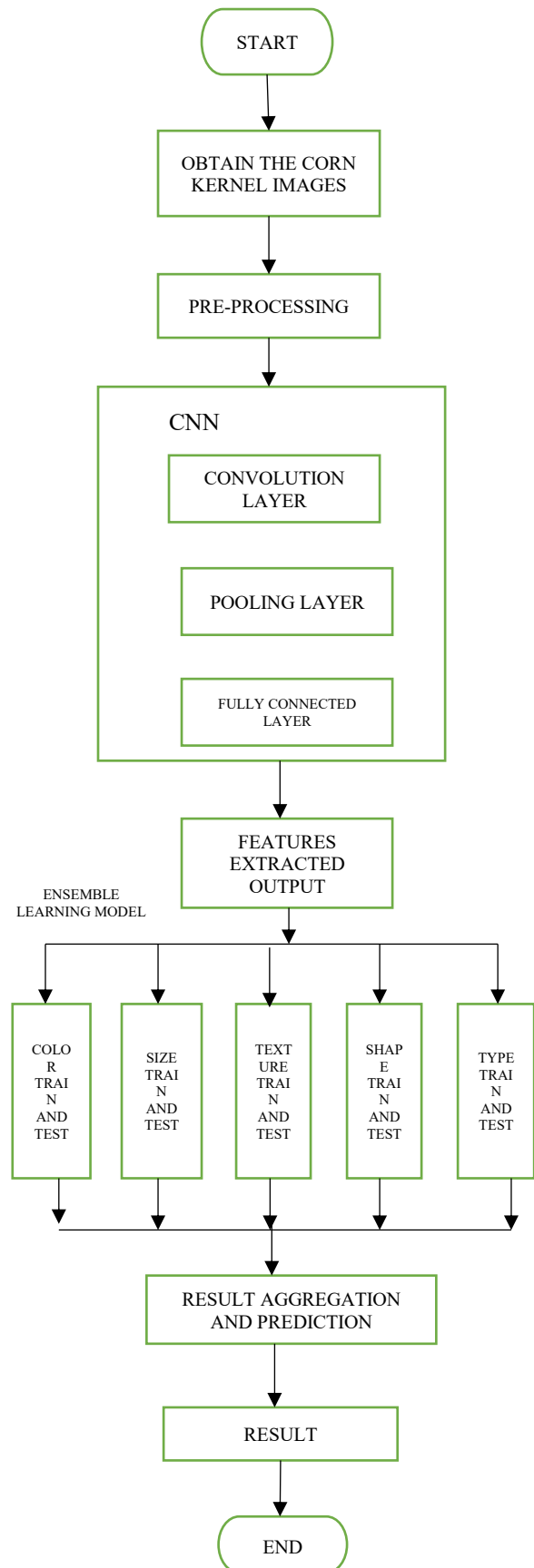


Fig.3. Flow chart of the proposed model

extracted output but not the image sets. Once the features are learned, tested and trained by the ensemble learning algorithm, with the help of final variance value, the corn kernel with minimum variance is selected. Here as the feature extraction is done using CNN and extracted features are tested using the ensemble learning that improves the prediction result of the proposed work. This work actually reduces the number of convolution layer that is compensated by the ensemble learning model that produces even more accurate results. As the number of convolution layer increases at times the process that inefficiently takes place. Proposed work reduces the complexity of the CNN and detects the corn kernel seeds of good quality.

V. RESULTS AND DISCUSSION

The proposed work is processed with the Corn Kernel Images dataset from Kaggle. It contains 78881 individual kernel image set that is photographed after dried, weighed and hand shelled. With the help of this dataset, the proposed work of utilizing CNN and ensemble learning algorithm is simulated. The CNN using five layer and ensemble learning uses five bagging for prediction.

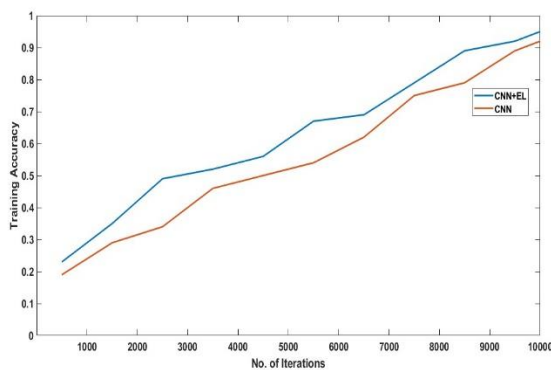


Fig. 4. Accuracy obtained for proposed work

Fig. 4. Shows the accuracy obtained for the proposed work of using CNN and Ensemble deep learning for the corn seed image classification. The proposal mainly concentrates for the number of datasets increases, the CNN requires more layer for feature extraction and sometime leads to unpredictable situation of more hidden neurons. As the incorporation of Ensemble learning algorithm greatly reduces the burden of CNN at attains more accuracy with normal convolutional layers of CNN.

From the figure it is clear that proposed approach accuracy is high compared to conventional method of using CNN for image classification. The accuracy seems to be 20 to 25% increase compared to the conventional method using CNN. Throughout the 10,000 iteration, the proposed work shown better performance comparatively.

The classification of good quality corn seeds by extracting the features using CNN and predicting the result using bagging ensemble learning algorithm efficiently classified the image. -

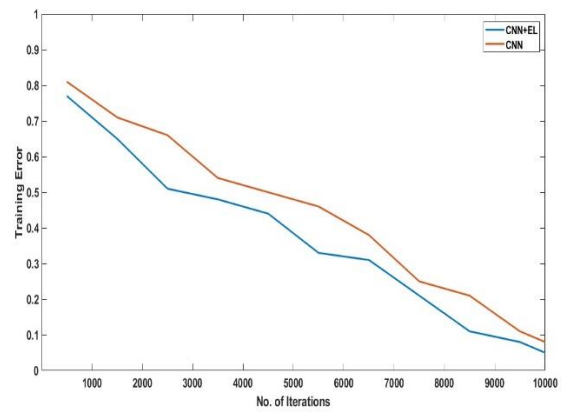


Fig.5. Error obtained for Proposed work

Fig. 5. Depicts the error occurred for the proposed work. As the iterations increases, the occurrence of error reduces, this is because, the image features are extracted by the convolutional layers in different iterations increases the accuracy and reduces the error occurring. The variance factor of ensemble learning helps the model to reduce error by choosing the image set containing the minimum variance value that indirectly decreases the error occurrences.

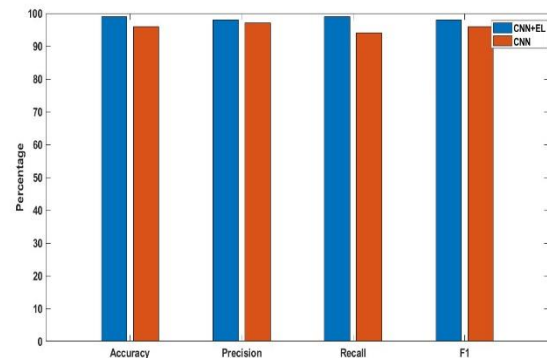


Fig. 6. Evaluation Metrics

Fig. 6. Shows the evaluation metrics for deep learning algorithm. For all four parameters, accuracy, precision, recall and F1, the proposed work shown better result compared to the conventional method of using CNN alone for image classification. The increase in the accuracy is mainly due to the incorporation of ensemble learning that predicts the corn seed based on the training and testing from five training and testing features. The precision has increased due to variance factor of the ensemble learning, which reduces the falseness completely. The recall efficiency obtained is due to many parameters involved in the feature prediction. Altogether it increased a tradeoff that results in the good F1 score.

VI. CONCLUSIONS

Thus the proposed work of using CNN and ensemble learning algorithm is used for image classification of corn seeds. As many of the food production industries demands good quality corn that mainly depends on the properties like shape, color, texture, size and type. This paper

proposes the novel technique of using ensemble learning after the feature extraction from CNN is trained with this five models and predicts the accurate output efficiently. The proposed work efficiently reduces the CNN burden for feature extraction for more amount of corn seed as it requires more layers. The future work of this work will extended by considering more physical properties to identify different standards of corn seeds for various purposes.

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